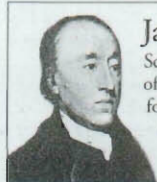


GEOLOGY

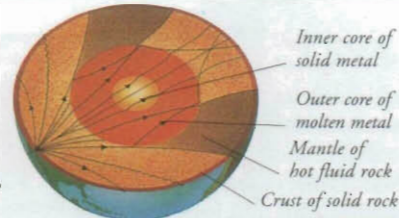


PEOPLE ONCE THOUGHT that the Earth was just a simple ball of rock. Recently geologists have shown that it is much more complex. Geology is the study of the Earth's history, structure, and composition. Originally only the study of rocks and rock structures, the scope of geological study broadened after the discovery in the 1960s that the Earth's crust is made up of giant, continually moving plates. These plates affect everything from the creation of continents to the eruption of volcanoes. The science of geology also helps us to locate mineral reserves, and to understand our environment.



James Hutton

Scottish-born James Hutton (1726–97) was the founder of modern geology. With his collection and analysis of rock formations, he proved that the Earth was more than just a few thousand years old, and that all its rocks and landforms had been formed over millions of years.



Structure of the Earth

Geophysicists and geochemists study the structure of the Earth. Geophysicists focus on its physical processes, such as the circulation of heat deep inside; geochemists study the Earth's chemical composition.

Seismographs

The seismic waves, earthquake vibrations, are picked up by seismographs. These can reveal to geologists the structure of the rock they have passed through.



Seismographs

Angular unconformity – the older rock strata below the unconformity are at a different angle to the new layers.

Rock and field geology

Petrology is the study of rocks and minerals. Surveys and rock samples indicate the occurrence of different rocks beneath the landscape, their structure, and their history.

The rock strata

Many rocks were formed in strata (layers) of sediment deposited on the seabed. Stratigraphy is the study of these layers. A break in a sequence of rock layers is called an unconformity, shown as a red line on the models below.

Limestone pavement

Parallel conformity – strata either side of unconformity dip at the same angle.

Rock strata models



Evidence of old landscape shaped long before the rocks above were formed.

Disconformity – an irregular eroded surface between parallel strata.

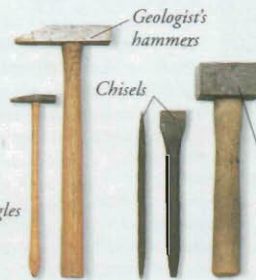
Nonconformity – strata overlie eroded surface of igneous or metamorphic rock

Key to strata:

Shale	Igneous rock
Conglomerate	Mudstone
Red Sandstone	Clay
Unconformity	Sandstone



Goggles



Geologist's hammers

Chisels

Tools

In order to examine the Earth's structure, geologists need some basic tools. These are goggles to protect their eyes from flying rock chips, and a hammer and chisel for collecting rock samples.

Club hammer for use with chisels.

Historical geology

The study of rocks of the Earth's crust is explored within historical geology. Just as the day is split into hours, minutes, and seconds, geological history is split into units called eras (lasting millions of years), periods, and epochs.

Geological rock column

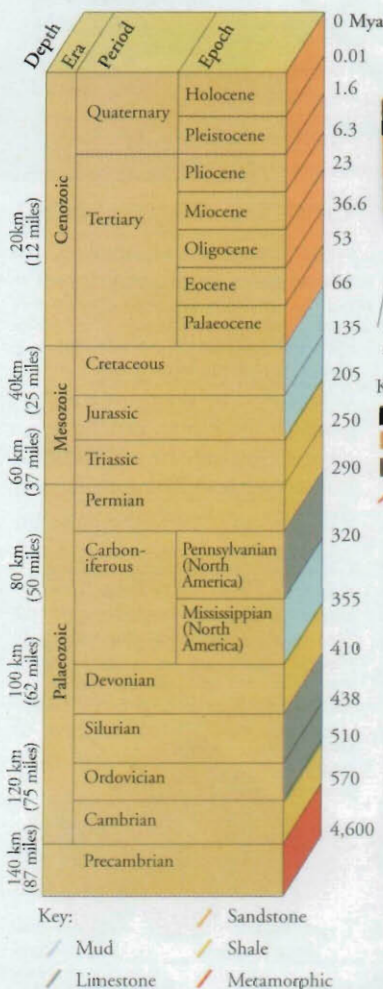
If layers of rock remained undisturbed, a column cut down through the layers would reveal the sequence in which they formed. The rock types shown along the side of this visual representation of geological time, are the predominant rocks of each period.

Geologists locate rock structures likely to contain oil; drilling can confirm this.



Exploration and survey

Each mineral in the Earth's crust is linked to a different type of geological structure. After using satellite and aerial surveys to target a particular area, the geologist then uses specific instruments to pinpoint the mineral.



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